

ASSIGNMENT NO. 9

PROBLEM STATEMENT:

Write a program in C to check whether the given number is an Armstrong number or not.

An Armstrong number is an integer with three digits such that the sum of the cubes of its digits is equal to the number itself.

Example: 371

OBJECTIVE:

- To understand the concept of Armstrong numbers.
- To practice arithmetic operations in C.
- To develop logical thinking using loops and conditional statements.

THEORY:

An Armstrong number (also called a narcissistic number) is a number that is equal to the sum of the cubes of its digits. For a three-digit number abc , it is an Armstrong number if:

$$a^3 + b^3 + c^3 = abc$$

For example, 371 is an Armstrong number because:

$$3^3 + 7^3 + 1^3 = 27 + 343 + 1 = 371.$$

PLATFORM:

Linux

CODE:

```
#include <stdio.h>
#include <math.h>

int main() {
    int num, originalNum, digit, sum = 0;

    printf("Enter a three-digit number: ");
    scanf("%d", &num);

    originalNum = num;

    // Extract each digit and calculate sum of cubes
    while (num > 0) {
        digit = num % 10;    // Get last digit
        sum += pow(digit, 3); // Add cube of digit to sum
        num /= 10;          // Remove last digit
    }

    // Check if sum equals original number
    if (sum == originalNum) {
        printf("%d is an Armstrong number.\n", originalNum);
    } else {
        printf("%d is not an Armstrong number.\n", originalNum);
    }

    return 0;
}
```

OUTPUT:

```
Enter a three-digit number: 371
371 is an Armstrong number.
```

CONCLUSION:

Thus, we have successfully written a C program to check whether a given number is an Armstrong number using loops and arithmetic operations.